



Major Depressive Disorder (MDD)

Introduction

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- **MDD**-is characterized by chronic feelings of sadness, guilt, discouraged, anxious or worthlessness without a history of manic or hypomanic disorder
- one or more major depressive episodes (i.e. at least 2 weeks of **depressed mood** or **loss of interest** accompanied by at least 5 symptoms of depression)
- **MDD Affects mood, thinking, physical health, work, and relationships**

Classification of disorders

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□ Depressive Disorders

- MDD, Single Episode or Recurrent

- Dysthymic Disorder

- Persistent, mild depressive disorder (2> yrs.)

□ Secondary Mood Disorder due to Non-psychiatric Medical Condition

□ Substance-Induced Mood Disorder

□ Mood Disorder Not Otherwise Specified

Etiology and Epidemiology

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- A combination of genetic, environmental, and psychological factors causes depression
- There is clear evidence that depression may occur as a result of **stressful events** (i.e., environmental factors) in one's life.
 - ▣ These factors include a **difficult childhood, physical or verbal abuse, pervasive low self-esteem, death of a loved one, loss of a job, and end of a serious relationship.**
- The annual incidence of mood disorders is approximately 10% in adult population

Risk factors for depression

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- Genetic predisposition (family hx)
 - ▣ **2.5X** greater in 1st degree relatives
- **Gender:** female two times than male
- **Age:** elderly are more affected
- **Race:** Whites > native Americans > African American
- Co-morbidity
- **Marital status:** single > divorced > widowed
- Substance abuse and medications

Cont.....

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- The presence of a significant **medical disorder** can produce depressive symptoms
 - ▣ Hypothyroidism, anemia, infections, electrolyte disturbances, CVD, neurologic disorders and **cancer**.
 - ▣ Various **psychiatric conditions**, such as substance-use disorders and anxiety disorders.

Selected Medications that May Induce Depression

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Cardiovascular Agents

β -blockers (?)

Clonidine

Methyldopa

Reserpine

Central Nervous System Agents

Barbiturates

Benzodiazepines (?)

Chloral hydrate

Ecstasy (MDMA)

Ethanol

Hormonal Agents

Anabolic steroids

Corticosteroids

Gonadotropin-releasing hormone

Progestins

Tamoxifen

Others

Efavirenz

Interferon

Isotretinoin

Mefloquine

Levetiracetam

Pathophysiology

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□ **Biogenic amine hypothesis**

✓ *Decreased brain levels of **NE, 5-HT** and **DA***

□ **Postsynaptic changes in receptor sensitivity**

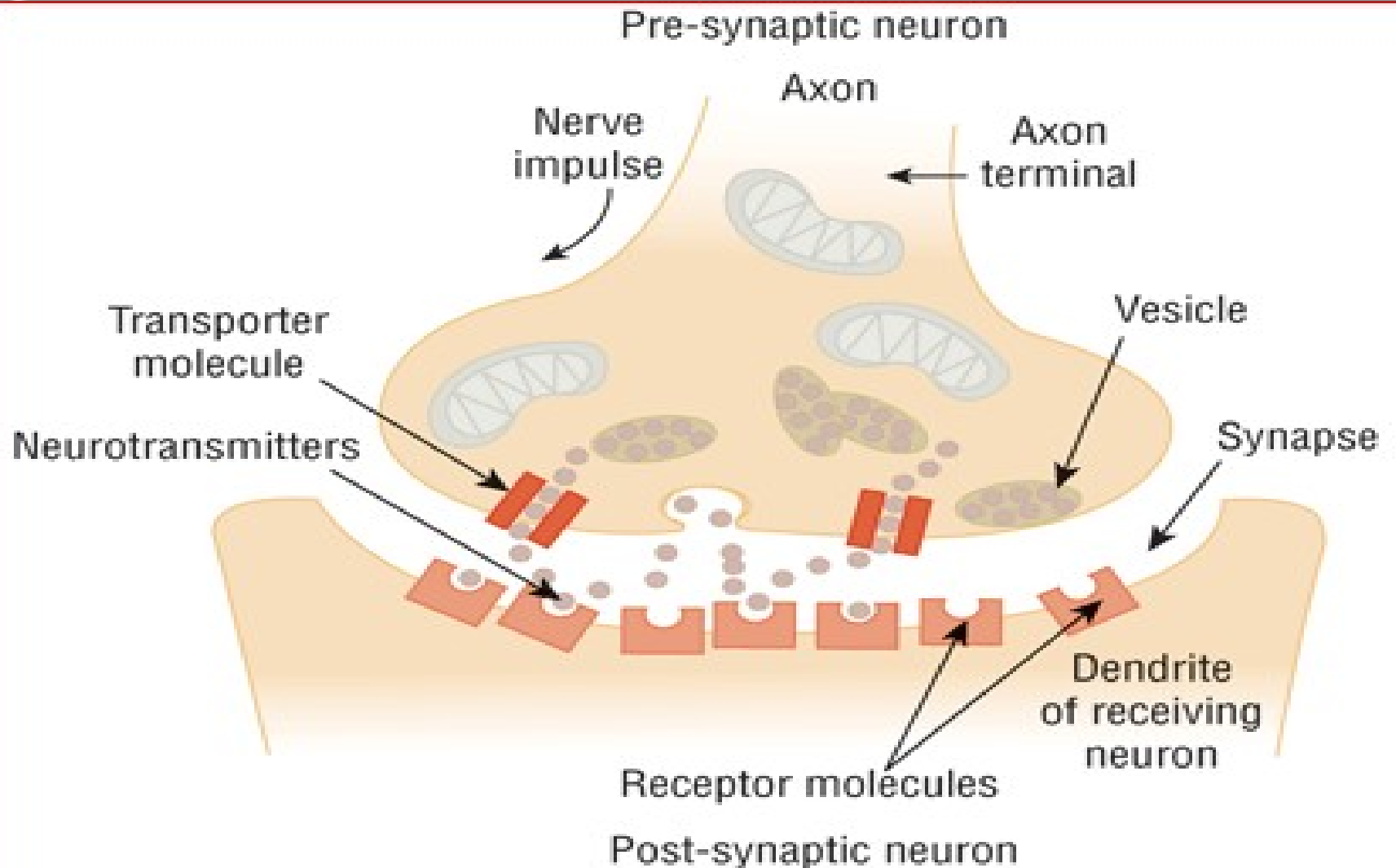
✓ *Desensitization or down regulation of **NE** or **5-HT1A** receptors*

□ **Dysregulation hypothesis**

□ *This theory emphasizes failure of homeostatic regulation of NT systems, rather than absolute increases or decreases in their activities.*

Monoamine NT regulation at neuronal level

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5-HT/NE link hypothesis

- ▣ *This theory suggests that there is a link between 5-HT & NE activity & that both serotonergic and noradrenergic systems are involved in the antidepressant response*

Role of DA

- ▣ *Several reviews suggest that **increased Dopamine in mesolimbic pathway** may be related to MOA of antidepressants*

□ **Emotional symptoms may include:**

- Diminished ability to experience pleasure,
- Loss of interest in usual activities, sadness, pessimistic outlook, crying, hopelessness, anxiety (present in almost 90% of depressed pts)
- Feelings of guilt & psychotic features (auditory hallucinations, delusions)

□ **Physical symptoms may include:**

- ◆ Fatigue, pain (especially headache), sleep disturbance, appetite disturbance (decreased or increased), loss of sexual interest
- ◆ GI and cardiovascular complaints (especially palpitations)

- **Intellectual or cognitive symptoms:**

- Impaired concentration or slowed thinking, poor memory for recent events, confusion & indecisiveness.

- **Psychomotor disturbances**

- Psychomotor retardation (slowed physical movements, thought processes & speech) or psychomotor agitation.

- In severe depression **psychotic symptoms**

Diagnostic Criteria: *A SAD FACES*

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- ▣ **A:** Appetite change (decrease/increase)
- ▣ **S :** Sleep disturbance (insomnia)
- ▣ **A:** Anhedonia (loss of interest in life)
- ▣ **D:** Depressed mood
- ▣ **F :** Fatigue
- ▣ **A:** Agitation
- ▣ **C :** Concentration loss
- ▣ **E:** Esteem (low self-esteem)
- ▣ **S:** suicide- Recurrent thoughts of suicide

DSM-IV Criteria for Major Depressive Episode

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- At least five of the following symptoms have been present during the same 2-week period and represent a change from previous functioning. One of the symptoms must be either depressed mood or loss of interest/pleasure.
 - Depressed mood most of the day, nearly every day
 - Loss of interest in pleasurable activities most of the day, nearly every day
 - Change in weight or appetite (increase or decrease) when not dieting
 - Insomnia or hypersomnia nearly every day
 - Fatigue or loss of energy nearly every day
 - Diminished ability to think or concentrate, or indecisiveness
 - Feelings of worthlessness or excessive or inappropriate guilt nearly every day
 - Psychomotor agitation or retardation nearly every day
 - Recurrent thoughts of death or suicidal ideation
- Symptoms cause clinically significant distress or impairment in social, occupational, or other important areas of functioning.
- Symptoms are not caused by an underlying medical condition or substance (e.g., medications or recreational drugs).

- Diagnosis requires a medication review, physical and mental status examination, CBC with differential, thyroid function tests, and electrolyte determinations.
- Many chronic illnesses and substance abuse and dependence disorders are associated with depression.
- Medications that can cause depression should also be reviewed.

Treatment

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□ Goals of therapy:

- Eliminate or reduce symptoms of depression
- Ensure compliance with therapeutic regimen
- Facilitate a return to a pre-morbid level of functioning
- Prevent further episodes of depression
- Minimize adverse effects of the drugs

Non-Pharmacologic Treatment

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Psychotherapy (talking with a psychiatrist, psychologist)

- The efficacy of **psychotherapy** is considered to be additive.
- Psychotherapy alone is not recommended for acute treatment of pts with severe and/or psychotic MDD
 - ▣ **Cognitive therapy** (focuses on present thinking, behavior, and communication rather than on past experiences and is oriented toward problem solving)
 - ▣ **Behavioral therapy** (identify and help change potentially self-destructive or unhealthy behaviors)
 - ▣ Interpersonal psychotherapy appear to be equal in efficacy

Electroconvulsive therapy (ECT)

- ▣ Is a safe and effective treatment for MDD but high risk of relapse
- ▣ **It is considered when**
 - Rapid response (***10-14 days***) is needed
 - Risk of other treatment outweighs potential benefits,
 - Depression in pregnancy
 - There has been a poor response to drugs, and
 - The pt expresses a preference for ECT

Pharmacological therapy

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- ◆ **Acute phase:** treatment lasts *6 to 10 wks* & the goal is **remission** (i.e., absence of symptoms)
- ◆ **Continuation phase:** lasts 4 to 9 mths after remission.
 - ✓ The goal is **to eliminate residual symptoms or prevent relapse.**
- ◆ **Maintenance phase:** lasts at least 12 to 36 mths & the goal is to **prevent recurrence** of a separate episode of depression.

Pharmacological therapy

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In general, antidepressants are equal effective if administered in comparable doses.

- Choice depends on
 - ▣ Patient's or family member's history of response,
 - ▣ Concurrent medical conditions
 - ▣ Presenting symptoms
 - ▣ Potential for drug–drug interactions
 - ▣ Side effect profiles,
 - ▣ Patient preference
 - ▣ Drug cost

Pharmacological therapy

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- Between 65% and 70% of patients with major depression improve with drug therapy.
- Psychotically depressed individuals generally require either ECT or combination therapy with an antidepressant and antipsychotic.

Educating the pts regarding:

- Delay in AD effects (typically 2 - 4 wks) &
- The importance of adherence should be emphasized before therapy is started & throughout treatment.

Pharmacologic Agents

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Antidepressant therapy

- ▣ Selective serotonin reuptake inhibitors (**SSRIs**)
- ▣ Serotonin norepinephrine reuptake inhibitors (**SNRIs**)
- ▣ Norepinephrine reuptake inhibitors (**NRIs**)
- ▣ Tricyclic antidepressants (**TCAs**)
- ▣ Monoamine oxidase inhibitors (**MAOIs**)
- ▣ Miscellaneous (trazodone, mirtazapine)

Pharmacology of Antidepressant Medications

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<i>Medication</i>	<i>Serotonin</i>	<i>Norepinephrine</i>	<i>Dopamine</i>	<i>Bioavailability (Oral)</i>	<i>Protein Binding</i>	<i>Half-Life (in hours) (Active Metabolite)</i>
Selective Serotonin Reuptake Inhibitors						
Fluoxetine (Prozac)	++++	0/+	0	80%	95%	24–72 (146)
Sertraline (Zoloft)	++++	0/+	+	>44%	95%	26 (66)
Paroxetine (Paxil)	++++	+	0	64%	99%	24
Citalopram (Celexa)	++++	0	0	80%	<80%	33
Escitalopram (Lexapro)	++++	0	0	80%	56%	27–32
Serotonin Norepinephrine Reuptake Inhibitors						
Venlafaxine (Effexor)	++++	+++	0	92%	25–29%	4 (10)
Duloxetine (Cymbalta)	++++	++++	0	50%	>90%	12 (8–17)
Norepinephrine Reuptake Inhibitors						
Bupropion (Wellbutrin)	0/+	+	+	>90%	85%	10–21
Tricyclic Antidepressants						
Desipramine (Norpramin)	+	++++	0/+	51%	90%	12–28
Nortriptyline (Pamelor)	++	+++	0	46%–56%	92%	18–56
Amitriptyline (Elavil)	++++	++++	0	37%–49%	95%	9–46 (18–56)
Imipramine (Tofranil)	+++	++	0/+	19%–35%	95%	6–28 (12–28)
Doxepin (Sinequan)	+++	+	0	17%–37%	68%–85%	11–23
Others						
Mirtazapine (Remeron)	+++	++++	0	50%	85%	20–40

0, negligible; +, very low; ++, low; +++, moderate; +++, high.

Selective serotonin reuptake inhibitors

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- SSRIs inhibit reuptake of 5-HT into the pre-synaptic neuron.
- Drugs and their usual dose
 - Citalopram 20–60 mg/QD
 - Fluvoxamine 50–300 mg/QD
 - Escitalopram 10–20 mg/day QD
 - Paroxetine 20–60 mg/day QD
 - Fluoxetine 20–60 mg/QD
 - Sertraline 50–200 mg/QD
- They are generally chosen as **first line drugs** because of their safety in overdose and improved tolerability compared to earlier agents.

SSRIs.....

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- SSRIs produce fewer sedative, anticholinergic, and cardiovascular adverse effects than TCAs
- The primary adverse effects are **nausea and vomiting, diarrhea, headache, insomnia, fatigue, and sexual dysfunction as well as weight gain**
- The effects of **SSRI on sexual function** are very common & potentially lead to medication **non-adherence**

SSRIs.....

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- SSRI-induced sexual dysfunction is approximately 30% to 50%, and it appears to be slightly more common in men, but the severity may be worse in women.
 - ▣ **Delayed orgasm** is the most common sexual complaint attributed to SSRIs or SNRIs
- This iatrogenic effect on orgasms has actually been used to clinical advantage in men **reporting premature ejaculation**

SSRIs.....

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Counselling tips for patients:

- ❖ It will take at least 2 -4 weeks to see the response, so continue taking your pills
- ❖ Take in morning because it causes insomnia
- ❖ They causes weight gain

Tricyclic Antidepressants

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- Inhibition of reuptake of **NE** and **5-HT**, block adrenergic, cholinergic, and histaminergic receptors.
- ***Tertiary amines***
 - **Amitriptyline 100-300 mg/d**
 - Doxepine 100-300 mg/d
 - Clomipramine 100-250 mg/d
 - **Imipramine 200-350 mg/d**
- ***Secondary amines***
 - Desipramine 100–300mg/d
 - Nortriptyline 50-200 mg/d

Side Effects

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□ Anticholinergic side effects

□ Dry mouth, blurred vision, constipation, urinary retention, tachycardia, memory impairment, delirium and **sedation** are more likely to occur with the 3⁰ amine TCAs.

□ **Orthostatic hypotension** and resultant syncope, occurs as a result of **α1-adrenergic antagonism**.

□ Cardiac conduction delays and heart block.

Side effects.....TCAs.....

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- **Weight gain** and **sexual dysfunction**.
- **Abrupt withdrawal of TCAs** (especially high doses) may result in symptoms of **cholinergic rebound** (dizziness, nausea, diarrhea, insomnia, restlessness).

Monoamine oxidase inhibitors

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- **Phenelzine, Selegiline and tranylcypromine** increase concentrations of NE, 5-HT, and DA.
 - ▣ Both non-selective inhibitors of MAO-A and MAO-B.
- **Selegiline** is available as a transdermal patch for treatment of major depression.
 - ▣ It inhibits MAO-A and MAO-B in the brain, but has **reduced effects on MAO-A in the gut.**

Side Effects

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- **Postural hypotension** (phenelzine than tranylcypromine)
- **Anticholinergic side effects**
- **Sedation** (by phenelzine)
- **Sexual dysfunction** in both genders is common.
- Hepatocellular damage and weight gain (Phenelzine).
- **Hypertensive crisis (when tyramine contain food is taken)**
 - ▣ It can be treated with captopril

Serotonin-NE Reuptake Inhibitors

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- Include **venlafaxine** and **duloxetine**.
- **Desvenlafaxine** 50 mg once daily.
- **Venlafaxine** may cause a dose-related **increase in diastolic BP**.
 - ▣ Dosage reduction or d/c may be necessary if HTN occur
- Other side effects: nausea and sexual dysfunction.

- **Maprotiline** and **amoxapine** are inhibitors of NE reuptake, with less effect on 5-HT reuptake.

Duloxetine

- The most common side effects are nausea, dry mouth, constipation, decreased appetite, insomnia, and increased sweating.

Miscellaneous

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- Trazodone and nefazodone inhibit the reuptake of 5-HT and are antagonists at the 5-HT₂ receptor **Side effects**
 - Minimal *anticholinergic* effects
 - Sedation, dizziness, and orthostatic hypotension are the most frequent dose limiting side effects.
 - **A black box warning:** life-threatening liver failure with nefazodone.

Cont.....

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Bupropion

- Potent blockade of **DA reuptake**; and lesser extent of NE

Side Effect

- ▣ **Seizures** dose related (hx of head trauma or CNS tumor).
- ▣ **Ceiling dose (450 mg/day)**--- incidence 0.4%
- **Other side effects** include nausea, vomiting, tremor, insomnia, dry mouth, and skin reactions.

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Mirtazapine

- Antagonism of central **pre-synaptic α_2 -adrenergic receptors.**
- It also antagonizes 5-HT₂ and 5-HT₃ receptors, histamine receptors
- Most common adverse effects are **somnolence, weight gain, dry mouth, and constipation.**

- The **SSRIs** are often selected as **first-choice antidepressants in elderly pts.**
- **Bupropion** and **venlafaxine** may also be chosen because of their milder **anti-cholinergic** and **less frequent cardiovascular side effects**
- **Fluoxetine** is the only antidepressant that is FDA approved for treatment of depression in pts <18 yrs of age

□ Black box warning

- FDA has established a link between antidepressant use and **suicidality** in children, adolescents, and young adults 18 to 24 years old.

Treatment resistant/refractory depression

- The current AD may be stopped, and a trial initiated with an agent of unrelated chemical structure
- **Augmented by**
 - ▣ lithium
 - ▣ **Atypical antipsychotics like** risperidone, olanzapine

TABLE 68-1

Adult Dosing Guidance for Currently Available Antidepressant Medications

Drug	Brand Name	Initial Dose (mg/day)	Usual Dosage Range (mg/day)	Comments (eg, Maximum Daily Dosage, Suggested Therapeutic Plasma Concentration) ^a
Selective Serotonin Reuptake Inhibitors (SSRIs)				
Citalopram	Celexa	20	20–40	Doses greater than 40 mg/day not recommended due to QT prolongation risk; maximum 20 mg/day for CYP2C19 poor metabolizers or coadministration with CYP2C19 inhibitors
Escitalopram	Lexapro	10	10–20	Maximum 20 mg/day; dose may be increased to maximum daily dose after at least 1 week if needed; 5 mg tablet available for unique circumstances
Fluoxetine	Prozac	20	20–60	Maximum 80 mg/day; dose may be increased in 20 mg increments; doses of 5 or 10 mg/day have been used as initial therapy; doses >20 mg/day may be given in a single daily dose or divided twice daily
Fluvoxamine	Luvox	50	50–300	Maximum 300 mg/day; daily doses >100 mg total dose should be divided twice daily, with the larger dose given at night
Paroxetine	Paxil	20	20–60	Maximum 50 mg/day (IR); titrate 10 mg/day increments weekly Maximum 62.5 mg/day (CR); titrate 12.5 mg/day increments weekly
Sertraline	Zoloft	50	50–200	Maximum 200 mg/day; titrate 25 mg/day increments weekly
Serotonin–Norepinephrine Reuptake Inhibitors (SNRIs)				
Newer-generation SNRIs				
Desvenlafaxine	Pristiq	50	50	Doses up to 400 mg/day have been studied; however, AEs are increased and no additional benefit has been shown at doses exceeding 50 mg/day
Duloxetine	Cymbalta	30	30–90	Maximum 120 mg/day (given once or twice daily); doses exceeding 60 mg/day not shown to provide increased efficacy for the treatment of MDD
Venlafaxine	Effexor	37.5–75	75–225	Maximum 375 mg/day (IR); maximum 225 mg/day (ER); may increase in increments up to 75 mg/day at a minimum of every 4 days. Dose reductions may be required if sustained hypertension occurs

(continued)

Drug	Brand Name	Initial Dose (mg/day)	Usual Dosage Range (mg/day)	Comments (e.g., Maximum Daily Dosage, Suggested Therapeutic Plasma Concentration) ^a
Tricyclic antidepressants (TCAs)				
43 Amitriptyline	Elavil	25	100–300	Maximum 300 mg/day for MDD; may be given as a single daily dose at bedtime or in divided doses throughout the day
				Therapeutic serum level 100–250 ng/mL (370–925 nmol/L); parent drug plus metabolite (ie, nortriptyline)
Desipramine	Norpramin	25	100–300	Maximum 300 mg/day Suggested therapeutic concentration range for combined imipramine + desipramine: 150–300 ng/mL (550–1100 nmol/L)
Doxepin	Sinequan	25	100–300	Maximum 300 mg/day; may be given in a single daily dose at bedtime (if tolerated) or in divided doses throughout the day; a single dose should not exceed 150 mg
Imipramine	Tofranil	25	100–300	Maximum 300 mg/day; may be given in a single daily dose at bedtime (if tolerated) or in divided doses throughout the day Suggested therapeutic concentration range for combined imipramine + desipramine: 150–300 ng/mL (550–1100 nmol/L)
Nortriptyline	Pamelor	25	50–150	Maximum 150 mg/day; total daily may be given as a single daily dose (if tolerated) or 25 mg doses given three to four times daily Therapeutic serum level 50–150 ng/mL (190–570 nmol/L)
Norepinephrine and Dopamine Reuptake Inhibitor (NDRI)				
Bupropion	Wellbutrin	150	150–300	Please see text for proper dosing, which can help decrease seizure risk Maximum 450 mg/day (IR, ER), 400 mg/day (SR); ER dosed once daily; SR dosed once or twice daily; IR may be dosed up to three times daily 09/25/2026

TABLE 68-2

Relative Potencies of Norepinephrine and Serotonin Reuptake Blockade and Selected Side Effect Profile of Antidepressants

	Reuptake Antagonism		Anticholinergic Effects	Sedation	Orthostatic Hypotension	Seizures ^a	Conduction Abnormalities ^a
	Norepinephrine	Serotonin					
Selective Serotonin Reuptake Inhibitors (SSRIs)							
Citalopram	0	++++	0	+	0	++	++
Escitalopram	0	++++	0	0	0	0	0
Fluoxetine	0	+++	0	0	0	++	0
Fluvoxamine	0	++++	0	+	0	++	0
Paroxetine	0	++++	+	+	0	++	0
Sertraline	0	++++	0	0	0	++	0
Serotonin–Norepinephrine Reuptake Inhibitors (SNRIs)							
Duloxetine ^b	++++	++++	+	0	+	0	0
Venlafaxine ^c and desvenlafaxine	++++	++++	+	+	0	++	+
Tricyclic Antidepressants (TCAs)							
Amitriptyline	++	++++	++++	++++	+++	+++	+++
Desipramine	++++	+	++	++	++	++	++
Doxepin	++	++	+++	++++	++	+++	++
Imipramine	+++	+++	+++	+++	++++	+++	+++
Nortriptyline	+++	++	++	++	+	++	++

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(continued)

	Reuptake Antagonism		Anticholinergic Effects	Sedation	Orthostatic Hypotension	Seizures ^a	Conduction Abnormalities ^a
	Norepinephrine	Serotonin					
Mixed Serotonergic (Mixed 5-HT)							
Nefazodone	0	++	0	+++	+++	++	+
Trazodone	0	++	0	++++	+++	++	+
Vilazodone	0	++++	0	+	0	++	0
Norepinephrine and Dopamine Reuptake Inhibitor (NDRI)							
Bupropion ^d	+	0	+	0	0	++++	+
Serotonin and α2-Receptor Antagonist							
Mirtazapine	0	0	+	++	++	0	+

TABLE 68–4		Adverse Drug Reactions and Monitoring Parameters Associated with Select New-Generation Antidepressants	
Drug	ADR(s)	Monitoring	Comments
Antidepressants from Each Pharmacologic Class			
Common to all antidepressants			
46	Suicidality	Behavioral changes Mental status	(U.S. boxed warning) for all antidepressants; caregivers should be alerted to monitor for acute changes in behavior
Selective Serotonin Reuptake Inhibitors (SSRIs)			
Common to all SSRIs			
	Anxiety or nervousness	Assess severity and impact on patient functioning and quality of life	Most prominent on initial treatment; generally subsides over time as antidepressant causes neurochemical adaptations
	Insomnia	Sleep patterns	Among SSRI class: fluoxetine may be more activating; fluvoxamine and paroxetine may be more sedating
	Nausea	Frequency and severity	
	Serotonin syndrome	Autonomic function (eg, pulse, temperature); neuromuscular function	Criteria include mental status change, clonus, hyperthermia, diaphoresis, and tachycardia
	Sexual dysfunction	Assess severity and impact on patient functioning and quality of life	Spontaneous self-reporting may be low; clinician should assess symptoms; reversible on drug discontinuation
SSRI-Specific			
Citalopram	QT interval prolongation	Electrocardiogram; electrolytes (eg, potassium, magnesium)	Caution use in “at-risk” patients (eg, electrolyte disturbance); discontinue if QTc persistently >500 milliseconds
Fluoxetine	Anorexia	Weight (over time)	SSRIs are generally considered weight neutral

Fluvoxamine	Somnolence	Mental status	May be less tolerable than other SSRIs
Paroxetine	Anticholinergic effects	Symptoms: dry mouth, constipation, urinary retention, mental status	Paroxetine possesses relatively more anticholinergic effects than other SSRIs

Serotonin–Norepinephrine Reuptake Inhibitors (SNRIs)

Common to all SNRIs

Insomnia	Sleep patterns	Possibly less likely with duloxetine
Nausea	Frequency and severity	
Serotonin syndrome	Autonomic function (eg, pulse temperature); neuromuscular function	Criteria include mental status changes, clonus, hyperthermia, diaphoresis, and tachycardia
Sexual dysfunction	Assess severity and impact on patient functioning and quality of life	Spontaneous self-reporting may be low; clinician should assess symptoms; reversible on drug discontinuation

SNRI-Specific

Desvenlafaxine	Hyperlipidemia	Lipid profile	Elevations in total cholesterol, low-density lipoproteins, and triglycerides
Duloxetine	Orthostatic <i>hypo</i> -tension	Blood pressure, pulse	Initial treatment or on dose increase
Venlafaxine	Dose-related <i>hyper</i> -tension	Blood pressure, pulse	May need to lower dose or discontinue

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TABLE 68-4

Adverse Drug Reactions and Monitoring Parameters Associated with Select New-Generation Antidepressants (Continued)

Drug	ADR(s)	Monitoring	Comments
Mixed Serotonergic Effects (Mixed 5-HT)			
⁴⁸ Nefazodone	Liver toxicity	Liver function tests	Nefazodone use is extremely limited in the United States due to concerns about liver toxicity
Trazodone	Orthostatic hypotension	Blood pressure, pulse	May be more severe as compared with other antidepressants; rate-limiting side effect
	Priapism	Patient report of sexual side effects, especially painful erection	Patient should seek medical attention for prolonged erection (ie, >4 hours)
Vilazodone	Serotonin syndrome	Autonomic function (eg, pulse temperature); neuromuscular function	Criteria include mental status changes, clonus, hyperthermia, diaphoresis, and tachycardia
Serotonin and α_2-Adrenergic Antagonist			
Mirtazapine	Weight gain	Body weight	Frequently occurring and significant (>7%) weight gain among adults
Norepinephrine and Dopamine Reuptake Inhibitor (NDRI)			
Bupropion	Seizure activity	Electroencephalogram	See text for proper dosing, which can help decrease seizure risk; caution use in patients with eating disorders or alcohol use disorders

Special population

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Elderly Patients

- ❑ **depressed mood** may be less prominent than other symptoms, such as loss of appetite, cognitive impairment, sleeplessness, fatigue, physical complaints, and loss of interest.
- ❑ The **SSRIs** are often considered first-choice antidepressants in elderly patients.
- ❑ **Bupropion, venlafaxine, and mirtazapine** are also effective and well tolerated.

- **Children**
- Symptoms of depression in childhood include anxiety, failing adjustment, and sleep disturbance.
- lack of data on efficacy of antidepressants in children and adolescents
- **Fluoxetine** is the only FDA approved antidepressant for treating depression in patients below 18 years of age
- Increased in **suicide** among children, adolescents, and young adults 18 to 24 years old.

Pregnancy

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- Generally, if effective, nondrug approaches are preferred
- If there is relapse after antidepressant discontinuation, it can be continued throughout pregnancy.
- There is a possible association of **SSRIs** with **low birth weight and respiratory distress**.
- **Six fold** greater likelihood of persistent **pulmonary htn** in newborns exposed to SSRIs after the 20th week of gestation.

- Consider the risks of untreated depression in pregnancy, including low birth weight, maternal suicidality, potential for hospitalization, poor prenatal care, and difficulty caring for other children.

Therapeutics Evaluation

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- Monitored for adverse effects, remission of target symptoms, and changes in social or occupational functioning.
- **venlafaxine BP** monitored regularly
- Monitor **suicidal ideation** for the first few weeks
- Psychometric rating instruments allow for rapid and reliable measurement

Case study

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- LM, a 40-year-old woman, presents to the inpatient psychiatric clinic after being referred from the mental health center. She is admitted secondary to stress over the holidays resulting in exacerbation of longstanding psychiatric illness. She complains of depressed mood, difficulty falling asleep, middle of the night awakening, early morning awakening, poor energy, poor concentration, recent weight loss of 6.8 kg over a 6-week period, nervousness, and frequent thoughts of death
- *What information is suggestive of MDD?*
- *What medical or psychiatric issues could be contributing to her symptoms?*
- *Does she have risk factors for depression?*

- *The goal of therapy for this patients?*
- *What non-pharmacologic and pharmacologic alternatives are available for this patient?*
- What are the monitoring parameters for the patients

The End!